

Songsoo Kim, MD, PhD

Higher degree

Yonsei University College of Medicine , Biomedical Systems Informatics (Supervisor: Dukyong Yoon)	Seoul, Republic of Korea 2026
Kyungpook National University, College of Medicine , Medicine	Daegu, Republic of Korea 2018
Yonsei University, College of Life Science and Biotechnology , Biochemistry	Seoul, Republic of Korea 2014

Postgraduate training

Department of Radiology, Seoul National University Hospital , Postdoctoral Research Fellow The Advanced Imaging and Computational Neuroimaging (AICON) Laboratory (PI&58 Kyu Sung Choi)	Seoul, Republic of Korea 2026 – present
Severance Hospital, Yonsei University College of Medicine , Research Fellow (Physician-Scientist Track)	Seoul, Republic of Korea 2023 – 2026
Department of Radiology, UMass Chan Medical School / UMass Memorial Medical Center , Visiting Research Assistant • Conducted research on Medical Ultrasound Artificial Intelligence with Prof. Young Kim, MD, PhD.	Worcester, MA, USA Oct 2022 – Oct 2022
Gangnam Severance Hospital, Yonsei University College of Medicine , Residency, Diagnostic Radiology • Chief resident (2022.07 – 2022.10)	Seoul, Republic of Korea 2019 – 2023
Yonsei University Health System , Internship	Seoul, Republic of Korea 2018 – 2019

Academic Service

- **Associate Editor**, *Radiology: In Training* — Radiological Society of North America (RSNA), 2026 – 2027.
- **Ad Hoc Peer Reviewer** — MICCAI 2026, PLOS Digital Health.

Awards

- **Trainee Research Prize** (Imaging Informatics category) — Radiological Society of North America (RSNA) (2025)
- **Merit Academic Paper Award** (Spring 2025) — Yonsei University Graduate School (2025)
- **Outstanding Research Achievement Scholarship** (Fall 2024) — Yonsei University College of Medicine (2024)
- **Outstanding Resident in Research** — Gangnam Severance Hospital, Yonsei University College of Medicine (2022)

Grants

- **European Congress of Radiology 2026 Travel Grant** — Korean Society of Radiology (KRW 4M / USD 3K), 2026
- **European Congress of Radiology 2025 Travel Grant** — Korean Society of Radiology (KRW 4M / USD 3K), 2025
- **Physician-Scientist Training Program** — Yonsei University College of Medicine (KRW 75M / USD 58K), 2023 – 2026
- **Full-time PhD Program**, MD-PhD/Medical Scientist Training Program — Korea Health Industry Development Institute, Ministry of Health & Welfare (KRW 225M / USD 173K), 2023 – 2026
- **Residency Program**, MD-PhD/Medical Scientist Training Program — Korea Health Industry Development Institute, Ministry of Health & Welfare (KRW 20M / USD 15K), 2022 – 2023

Publications

- See the full list on the [publications page](#).

Conference Activities

- **Oral Presentation**, *Abdominal CT and Brain MRI Phenotypes Across the Glycemic Spectrum: Graded Body-Brain Signatures of Prediabetes and Diabetes*, RSNA 2026, Chicago.
- **Poster Presentation**, *Agentic AI for Longitudinal Assessment of Brain Metastases: Reproducible Reporting and Human-in-the-Loop Safety*, RSNA 2026, Chicago.
- **Educational Poster (presenter)**, *AI-Driven Opportunistic Screening for Hidden Clinical Risks: A Practical Research Primer*, ECR 2026, Vienna — *Top-rated Poster, EPOS™ PULSE Live presentation*.
- **Digital presentation (presenter)**, *Opportunistic Screening of Diabetes Using AI-Extracted Whole-body MRI Biomarkers*, RSNA 2025, Chicago — *Trainee Research Prize*.
- **Oral presentation (presenter)**, *AI-Driven MRI Biomarker Extraction and Machine Learning Analysis of Their Association with Diabetes: A UK Biobank Study*, ECR 2025, Vienna.
- **Educational Poster**, *Do not overlook Hirayama disease: Even without a characteristic flexion MRI finding*, ECR 2025, Vienna.
- **Oral Presentation (presenter)**, *Leveraging GPT-4 for Error Detection on Brain CT Report: A Comparative Study of Different Report Formats and Error Types*, RSNA 2023, Chicago.
- **Oral Presentation (presenter)**, *Few-Shot Prompting for Labeling Free-Text Radiology Report Using GPT-4*, RSNA 2023, Chicago.
- **Oral Presentation (presenter)**, *Risk factors for spontaneous hemorrhage in brain metastases in patients with lung adenocarcinoma*, AOCR & KCR 2022, Seoul.
- **Oral Presentation (presenter)**, *Impaired meningeal lymphatic drainage in patients with mild cognitive impairment*, AOCR & KCR 2022, Seoul.
- **Organizing Committee Member**, Young Radiologists Forum, KSSR Spring Symposium 2022, Busan.

Invited Lectures and Seminars

- *Leveraging Large Language Models for Error Detection in Radiology Reports* — **2025 Spring Conference of the Korean Society of Medical and Biological Engineering (KOSOMBE)**, Clinician-Engineer Session, Jeju (2025.05.08).
- *Medical Data-Specific Quality Management Metrics* — **2024 Symposium of the Institute for Innovation in Digital Healthcare (IIDH)**, LYDUS Session, Severance Hospital, Seoul (2024.11.20).
- *Applying Large Language Models in Medicine with Python — A Practical Guide* — **Lab of Artificial Intelligence in Medicine, Graduate School of Medicine, Ajou University**, Suwon (2024.05.13).
- *Bulk Processing and Utilization of Large Language Models via APIs — Hands-On Training* — **2024 Winter School of the Korean Society of Medical Informatics (KOSMI)**, Seoul (2024.01.17).

Licenses and Certifications

- **Board of Radiology** (No. 4615), Ministry of Health & Welfare, Republic of Korea (2023)
- **Medical Doctor License** (No. 130233), Ministry of Health & Welfare, Republic of Korea (2018)

Professional development

- **Emergency Certificate**, Radiological Society of North America (RSNA) Imaging AI Certificate Program (2024)
- **Clinical Research Methodology Course (CRMC) – Advanced Program**, The Korean Society of Radiology (2023) — Completed a 2-day mentoring workshop on advanced research methodologies, including statistical techniques, and delivered a presentation on a selected topic.
- **Advanced Certificate**, Radiological Society of North America (RSNA) Imaging AI Certificate Program (2023)

Projects

Development of a Multidimensional Quality Evaluation System for Health and Medical Big Data

2023 – 2025

Government-funded national R&D project on medical data quality management, led by Yonsei University College of Medicine (PI: Prof. Dukyong Yoon) with Seoul National University Hospital, Samsung Medical Center, and Severance Hospital.

- Contributed as research associate, focusing on developing metrics, programs, and guidelines for medical data quality management.
- Presented project outcomes at consortium meetings.

Patents

- Yoon, D., **Kim, S.**, Ryu, J., Choi, Y., Han, C. *Apparatus and method for evaluating the quality of medical data based on fidelity*. Korean Patent Application No. 10-2024-0094441, filed July 17, 2024. Patent pending.
- Yoon, D., Han, C., Bae, S., **Kim, S.**, Kim, D. *Method for providing information on the risk of developing cardiovascular diseases and device using the same*. Korean Patent Application No. 10-2024-0091928, filed July 11, 2024. Patent pending.
- Yoon, D., Han, C., Ryu, J., Choi, Y., **Kim, S.** *Apparatus and method for evaluating the quality of medical data based on reliability*. Korean Patent Application No. 10-2024-0089258, filed July 5, 2024. Patent pending.
- Yoon, D., Choi, Y., Ryu, J., **Kim, S.**, Han, C. *Apparatus and method for evaluating the quality of medical data based on data consistency*. Korean Patent Application No. 10-2024-0050332, filed April 15, 2024. Patent pending.
- Yoon, D., Ryu, J., Han, C., **Kim, S.**, Choi, Y. *Apparatus and method for evaluating the quality of medical data based on precedence and precedence relationships*. Korean Patent Application No. 10-2024-0046975, filed April 5, 2024. Patent pending.